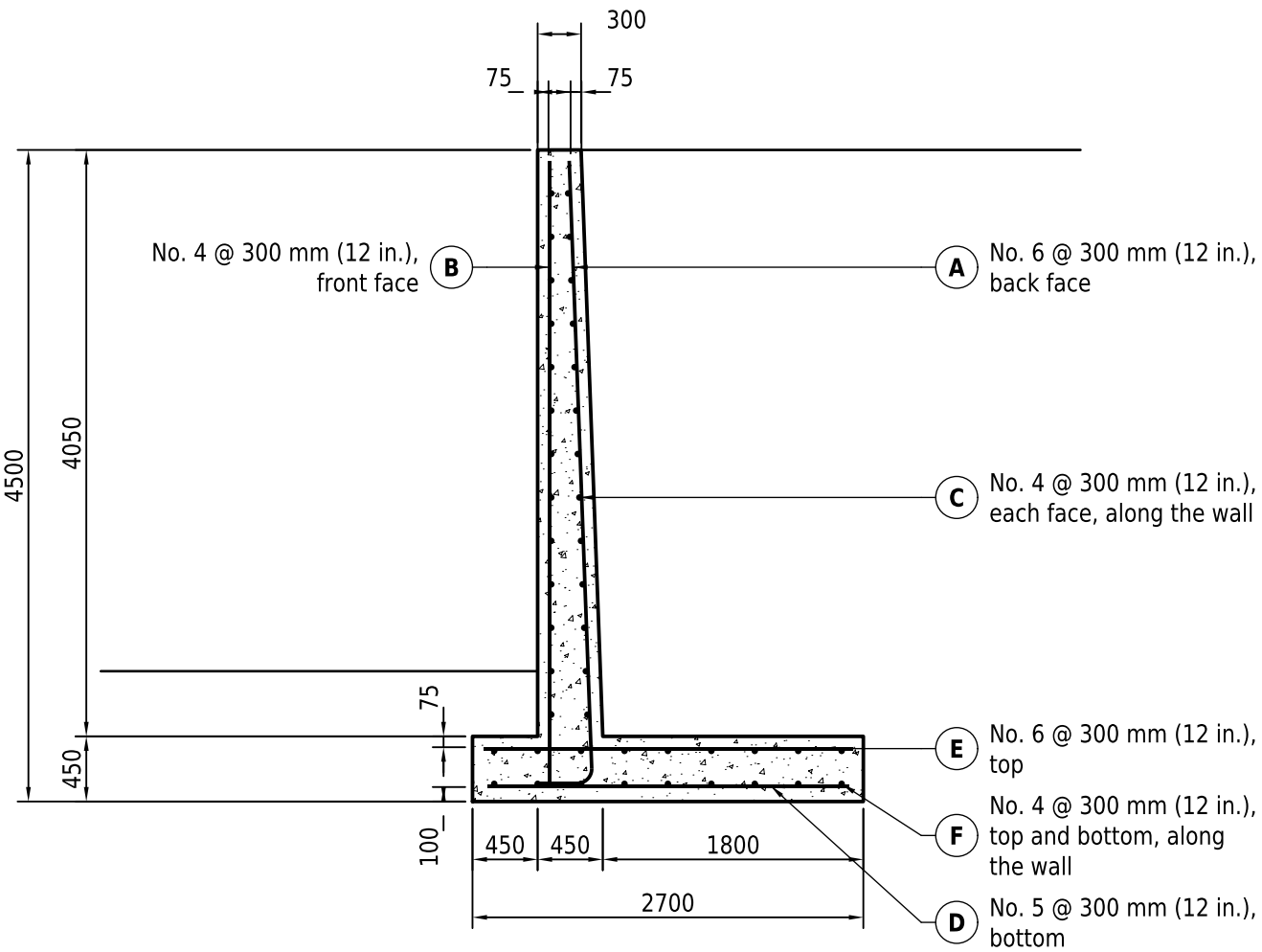
CALCULATION SKETCH - NOT FOR CONSTRUCTION



BAR SCHEDULE (steel areas in mm2 per metre of wall; along = bars along the wall)

Mark	Size	Spacing	Place	As req.	As prov.
A	No. 6	300 mm	Stem, back face	880	947
B	No. 4	300 mm	Stem, front face	400	430
C	No. 4	300 mm	Stem, each face, along	400	430
D	No. 5	300 mm	Toe, bottom	610	663
E	No. 6	300 mm	Heel, top	900	947
F	No. 4	300 mm	Base, top and bottom, along	400	430

SCALE 1:50

0 1 2 m

NOTES

1. Concrete f'c = 28 MPa (4,060 psi). Reinforcement fy = 414 MPa (60,000 psi).
2. Lap and straight development lengths and bar cut-offs are not designed.
3. Design basis: USACE EM 1110-2-2104 (1 November 2023), strength design with the serviceability single load factors
4. All dimensions are in millimetres.
5. Bars are drawn full length: no bar is cut off or lapped in this sketch. Bars along the wall are drawn end-on.
6. Hooked stem bars are turned toward the opposite face of the stem (EM 1110-2-2104 Fig. 2-1, correct orientation). The hook's bend and tail are schematic, not detailed.

Cantilever retaining wall: reinforcement (made-up example)

PROJECT: Made-up example	
NO.:	DWG: wall_example
DRAWN: GSE	CHECKED:
DATE: 2026-11-13	SCALE 1:50 (letter)